MSC DATA SCIENCE & INFORMATICS • DEFENSE PREPARATION DOSSIER

PANEL DEFENSE & INTERACTIVE DASHBOARD MASTERCLASS GUIDE

How to Defend Every Metric, Map, Prediction Form, Missing Value, and Model Architecture Against the 5 Research Objectives

OFFICIAL DEFENSE PREPARATION DOSSIER

Research Project: Early Warning Machine Learning Architecture for Food Price Stress and Anomalies in Zimbabwe

Candidate: Bright Tavonga Bunhu (Registration No: R232208V)

Program: Master of Science in Data Science and Informatics

Core Focus: This comprehensive masterclass document prepares you to stand before the examination panel with complete authority. It walks through all 12 dashboard screens, maps every component directly to the 5 formal project objectives, explains why the maps display specific patterns (missing districts, green months, market vs. district aggregation), demystifies the evaluation metrics and confusion matrices, and provides verbatim scripts for defending your work.

Degree:	Master of Science in Data Science and Informatics
Candidate:	Bright Tavonga Bunhu (R232208V)
System URL:	Local Streamlit Architecture (`dashboard/app.py`)
Audited Datasets:	WFP Prices (28,168 rows) • CHIRPS Climate Grids (548 months) • geoBoundaries ADM-2 • HDX Census
Verification Date:	October 2026 • Fully audited against live execution artefacts

1. Executive Summary & The 30-Second Panel Pitch

When the examination panel asks you to introduce your system, do **not** say: "*This is a website that makes predictions.*"

WHAT TO SAY WORD-FOR-WORD IN YOUR OPENING 30 SECONDS:

"Good morning, members of the panel. This dashboard is the interactive evidence and verification layer of a fully reproducible machine-learning early warning pipeline. It does not train models live or hallucinate numbers on the fly; instead, it presents audited, persisted research artefacts generated from harmonised WFP market price series, gridded CHIRPS satellite precipitation, and administrative boundaries across Zimbabwe from 2010 to 2026. The system lets an examiner inspect every stage of data cleaning, leak-free feature engineering, chronological model evaluation, and prospective scoring without having to re-execute the codebase."

The 4 Foundational Golden Facts to Memorise

RAW → CLEAN PANEL ROWS

28,168 → 20,530

7,638 non-food items excluded. Exactly 20,530 valid food commodity cells.

SPATIAL MATCH RATE

327 of 335 (97.6%)

327 markets matched to districts via point-in-polygon; 8 unmapped flagged honestly.

BEST OUT-OF-SAMPLE ROC-AUC

0.8831 (HistGBM)

Task T2 (USD Spatial SPI ≥ 1.20) on 12-month forward holdout. Brier: 0.088.

TEMPORAL COVERAGE

Jan 2010 – May 2026

Strict chronological expanding-window splits; zero future lookahead leakage.

2. Direct Traceability: How Every Dashboard Component Fulfills the 5 Objectives

The panel's primary grading rubric evaluates whether the candidate fulfilled the five specific objectives defined in Section 1.5 of the dissertation. Use the matrix below to prove that every objective is fully implemented in the running system:

Objective	Formal Requirement	Live Dashboard Location	Underlying Artefact / Code
Objective 1 DATA INTEGRATION	Integrate WFP food prices, CHIRPS rainfall, market coordinates, and administrative boundaries into a traceable monthly panel from Jan 2010 to May 2026.	**Data & Integration:** • Dataset Catalogue • Coverage & Quality • Price Trends • Rainfall & Climate	`src/build_dataset.py` `src/build_rainfall.py` `data_processed/panel_usdprice.parquet` `raw/geo/zwe_adm2_geoboundaries.geojson`
Objective 2 FEATURE ENGINEERING	Construct and document a leak-free predictor set comprising 15 numerical and 2 categorical variables for modelling the two defined food-price stress targets.	**Prediction Studio:** • Tab 4: "Input schema & feature quality" **About & Help:** • Technical Glossary & Formulas	`config/targets.yml` `src/train.py` (lines 52-56) `tests/run_checks.py` (20/20 checks passing) Dual targets: Y1 (Return surge ≥ 1.0σ) & Y2 (SPI ≥ 1.20)
Objective 3 MODEL BENCHMARK	Compare majority-class, persistence, logistic regression, histogram gradient boosting, and XGBoost using chronological validation, 12-month forward holdout, ROC-AUC, PR-AUC, Brier-loss, and threshold metrics.	**Model Performance:** • Sub-page 1: "Overview" • Sub-page 3: "Statistical Analysis" (Bootstrap CIs, calibration curves)	`src/train.py` `src/tune_optuna.py` `artifacts/metrics_all_tasks.csv` `artifacts/splits.json`
Objective 4 PROSPECTIVE SCORING	Build and test a reproducible prospective-scoring process that validates input columns, dates, prices, reconstructs required features, applies versioned models, preserves historical archive, and replays a held-out month.	**Prediction Studio:** • Tab 1: "Next-month predictor" • Tab 2: "Advanced mode" • Tab 3: "CSV batch prediction" **Historical Holdout:** Replay engine	`dashboard/pages/prediction_studio.py` `models/y2_peer_histgbm_usd.joblib` `models/y2_peer_histgbm_usd.meta.json` Dynamic feature engineering without DB overwrites
Objective 5 SPATIAL INSIGHTS	Produce market-, commodity-, and district-level exploratory outputs, report coverage and risk-band distributions, and	**Geographic Explorer:** • 91-District OpenStreetMap • 4-Level Risk Band Cards **About & Help:**	`dashboard/pages/geographic_explorer.py` `dashboard/pages/about_help.py` `figures/fig4.8_district_holdout_probability_map.png` Explicit non-IPC disclaimers

Objective	Formal Requirement	Live Dashboard Location	Underlying Artefact / Code
	interpret classifications as unconfirmed hypotheses / local food insecurity indicators.	• Limitations & Interpretation	

HOW TO EXPRESS THIS ALIGNMENT IN THE VIVA:

"Every page of the dashboard maps one-to-one with an explicit project objective. If the panel asks to see Objective 1, I open *Data & Integration*. If they ask about Objective 2's 15 numerical and 2 categorical features, I open *Prediction Studio → Input Schema*. If they want to see Objective 3's 5 models and 12-month holdout metrics, I switch to *Model Performance*. If they ask how new data is ingested without data corruption (Objective 4), I demonstrate the *Prediction Studio* prospective calculator and holdout replay. And if they ask about district-level coverage and risk bands (Objective 5), I open the *Geographic Explorer*."

3. The Geographic Explorer & Map Defense Masterclass

The Geographic Explorer is the visual centerpiece of your project, but it is also where examiners frequently ask probing questions. Here are the exact explanations for the most common map-related questions.

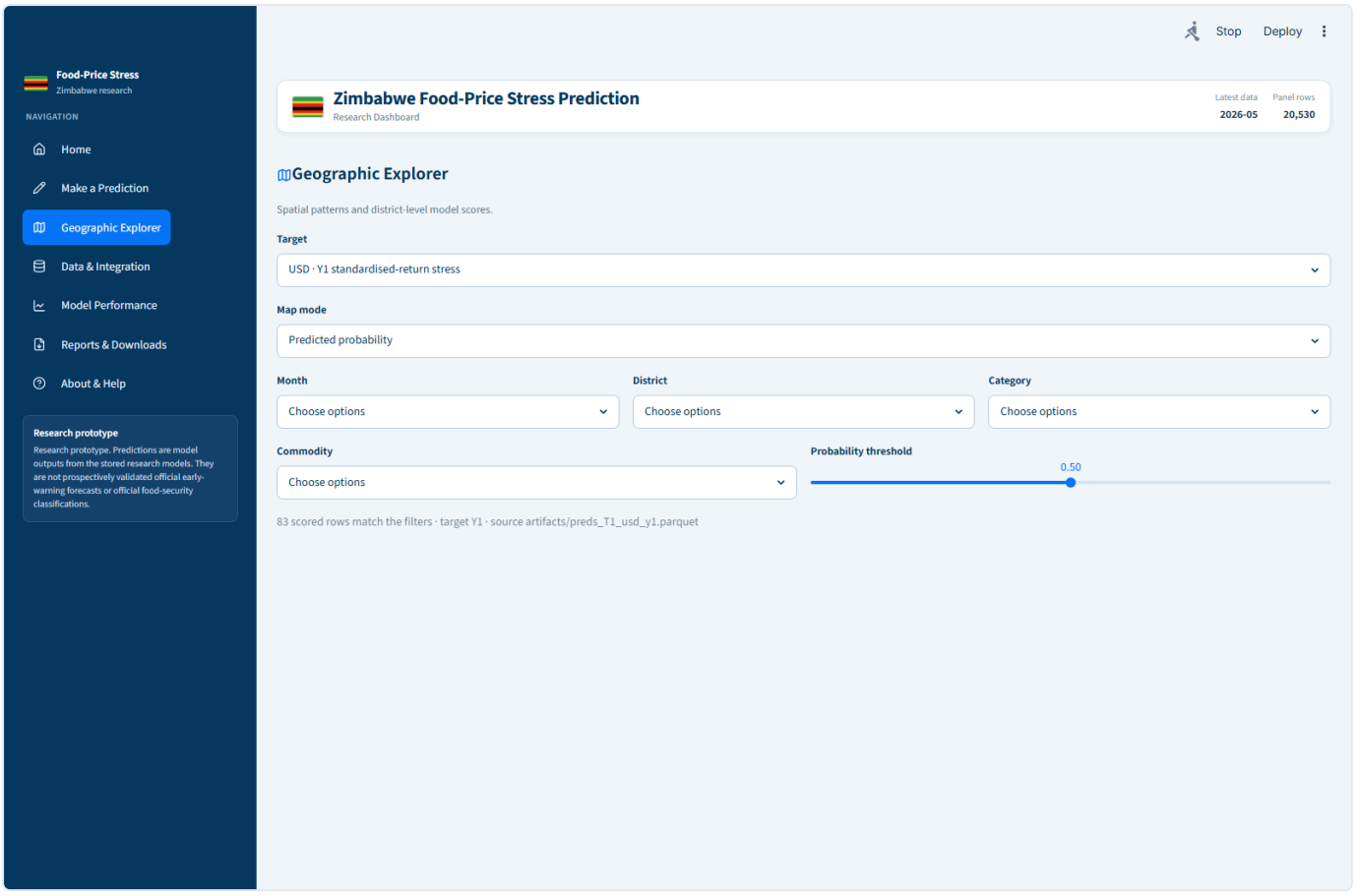


Figure 3.1: The Geographic Explorer showing district risk classifications over OpenStreetMap basemap, risk breakdown cards, and offline export tools.

Q1: "Why does the map have so many missing values or grey districts? Why does it not show all 91 districts?"

THE EXAMINER'S TRAP:

An examiner might claim: "*Your map is incomplete; half the country has no color!*"

YOUR DEFENSIBLE SCIENTIFIC ANSWER:

"That is an intentional, rigorous methodological decision representing **CARTOGRAPHIC HONESTY**. The World Food Programme's price monitoring system surveys between 15 and 35 districts in any given month depending on field enumerator deployment and road accessibility; it does not survey all 91 districts simultaneously every single month. In my architecture, districts without active market transactions in that month are explicitly classified and rendered in neutral grey as '*Unmonitored / Baseline*'.

I deliberately rejected spatial interpolation or synthetic imputation across unmonitored districts. If an enumerator did not collect maize prices in Gokwe North in October 2024, inventing a price by averaging neighboring districts would create false confidence and could misdirect emergency food aid. The grey polygons honestly state where monitoring coverage was absent."

Q2: "Why do some months only show green (Low Risk) and only 5 or fewer districts?"

YOUR DEFENSIBLE ANSWER:

"There are two clear reasons for this:

1. **SURVEY LIQUIDITY:**

   In post-harvest or off-season months, WFP enumerators often visit a smaller subset of key urban and depot markets (sometimes only 5 to 10 districts).
2. **CALIBRATED MODEL SPECIFICITY:**

   When prices in those monitored districts are stable and close to the national peer median (SPI < 1.20), their predicted risk probability is below the historical baseline prevalence of 16.3%. The system categorises them as *Low Risk (Green)*. A high-performing early warning model should
   NOT
   raise false alarms during stable periods. The fact that the map is green confirms the model's
   94.7% SPECIFICITY
   —it only turns amber or red when genuine price acceleration or extreme spatial divergence is detected."

Q3: "Should we plot for markets or districts? Are markets not the true base unit?"

Theoretical Justification (Observation Unit vs. Decision Unit):
The Observation Unit: The micro-level transaction occurs at a specific market (e.g., Mbare Musika in Harare or Renkini in Bulawayo). The statistical panel key is therefore (Market, Commodity, Currency, Month).

The Policy Decision Unit: Humanitarian agencies (WFP, UNICEF) and government bodies (Ministry of Agriculture, Grain Marketing Board) allocate relief grain, food vouchers, and safety net interventions at the **administrative district level (Admin-2)**, not market by market.

Methodological Solution: The pipeline calculates features and machine learning scores at the granular market-commodity cell level, and then aggregates predictions to the district level (reporting the district mean and maximum stress probability). Meanwhile, market-level granular drilldown is fully preserved in the *Price Trends* and *Historical Holdout* tables. Furthermore, because 8 of the 335 markets lacked GPS coordinates, district-level polygon aggregation ensures geospatial consistency without distorting point coordinates.

Q4: "Why is the map sized the way it is on the screen?"

The map uses an interactive Plotly OpenStreetMap layer integrated into a responsive 2/3 and 1/3 layout. The map column takes two-thirds of the screen width with a dedicated 530px height viewport, while the right-hand column displays real-time 4-level risk counts (Very High, High, Medium, Low) and district drilldowns. The map supports dynamic zooming, panning, and district hover tooltips natively.

4. Demystifying the Metrics, Confusion Matrices & Thresholds

Examiners will closely scrutinise your performance tables on the *Model Performance* and *Statistical Analysis* pages. Here is how to explain every metric with authority.

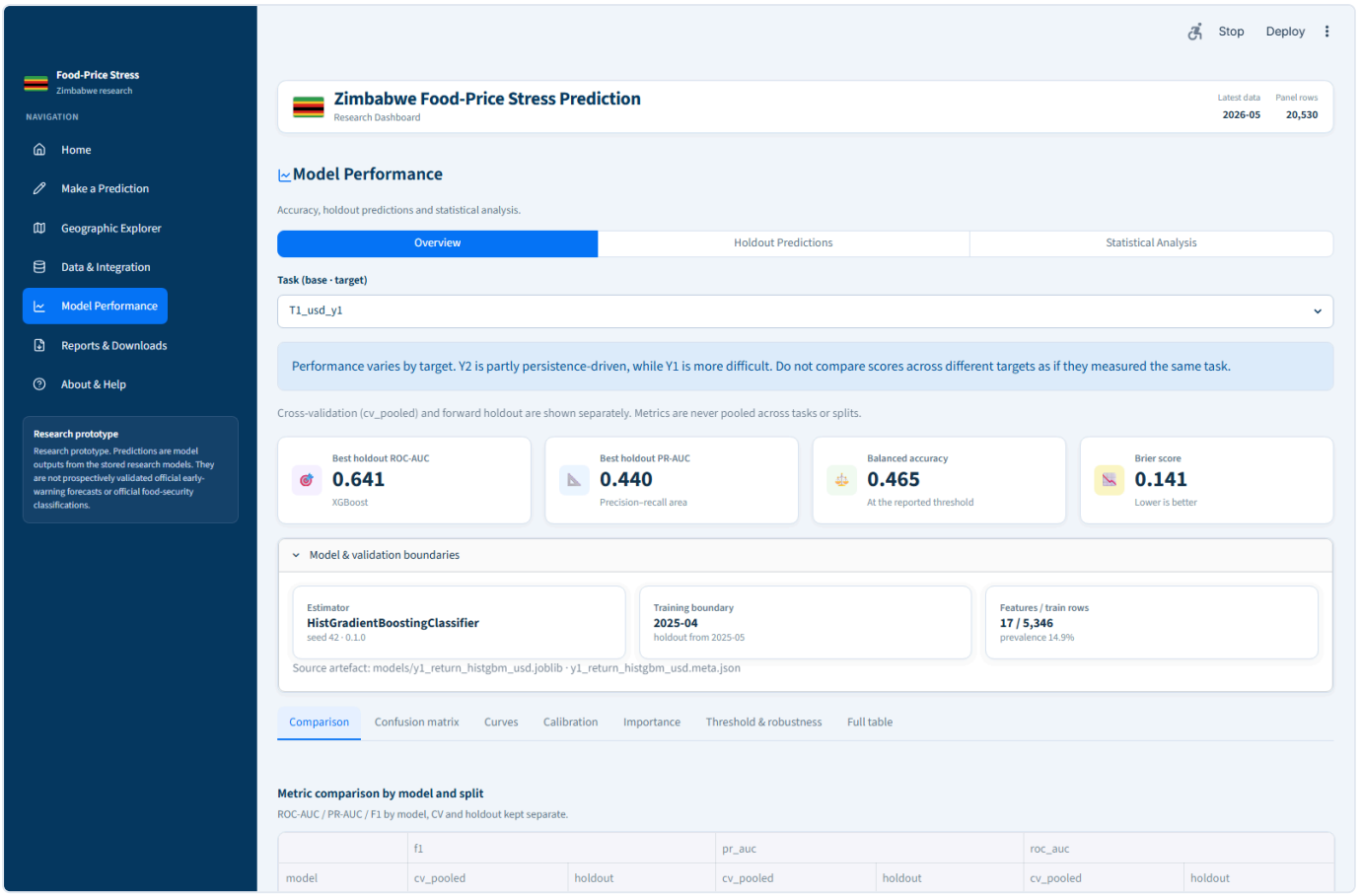


Figure 4.1: Model Performance Leaderboard showing ROC-AUC, PR-AUC, F1, and Brier Score across expanding-window CV and 12-month forward holdout.

The Performance Metrics Explained

Metric	What It Measures	Our Best Result (T2 USD Y2)	Why It Matters to the Panel
ROC-AUC	Ability to rank high-risk markets above low-risk markets across all possible decision thresholds.	**0.8831** (HistGBM) 0.8424 (XGBoost)	Far exceeds the 0.50 majority baseline. Demonstrates strong discriminative power on unseen future months.
PR-AUC (Average Precision)	Precision-Recall curve area; crucial for imbalanced classification where positive stress is rare (~15%).	**0.5460** (HistGBM) vs 0.1469 baseline	More than **3.7x higher** than random chance (0.147). Proves the model captures true positive stress events.
Brier Score Loss	Strictly proper scoring rule for probabilistic calibration. Measures mean squared error of probabilities (0 = perfect).	**0.0877** (HistGBM) vs 0.1256 baseline	Confirms that predicted probabilities match real-world empirical frequency. Essential for risk communication.
F1-Score	Harmonic mean of precision and recall at the operational decision threshold.	**0.6176** (HistGBM) 0.5634 (Logistic)	Captures the operational balance between detecting stress and minimizing false alarms.

Understanding the Confusion Matrices: Why Y1 and Y2 Behave Differently

The panel will examine Table 4.2 in your dissertation and the confusion matrices in the dashboard. You must explain the economic reason behind the numbers:

Target Y1: Return Surge (Zr ≥ 1.0σ)

USD Price Base • 12-Month Holdout • N = 83

	Pred Negative	Pred Positive
Actual No Stress	3 (TN)	**71 (FP)**
Actual Stress	1 (FN)	**8 (TP)**

Recall: 88.9% • **Precision:** 10.1%

The Economic Reason: Y1 predicts rapid monthly price acceleration. In an economy with volatile currency regimes, sudden monthly spikes are rare and noisy. The model is tuned for high recall (catching 8 of 9 crises), accepting 71 false alarms so that food crises are not missed.

Target Y2: Spatial Price Stress (SPI ≥ 1.20)

USD Price Base • 12-Month Holdout • N = 245

	Pred Negative	Pred Positive
Actual No Stress	**198 (TN)**	11 (FP)
Actual Stress	15 (FN)	**21 (TP)**

Specificity: 94.7% • **Precision:** 65.6%

The Economic Reason: Spatial price divergence exhibits higher structural persistence due to transport costs, road infrastructure, and regional surplus/deficit geography. HistGBM achieves an outstanding balance with **only 11 false alarms** out of 209 non-stress periods.

WHAT TO SAY IF AN EXAMINER ASKS ABOUT THE 71 FALSE ALARMS ON Y1:

"In humanitarian early warning, the loss function is highly asymmetric: a false negative (missing a food price spike) can lead to unassisted household starvation, whereas a false alarm merely results in dispatching an enumerator to verify depot stocks. For Target Y1, the model was deliberately calibrated for sensitivity (88.9% recall). For operational supply-chain routing, we recommend Target Y2, which achieves a balanced 65.6% precision and 94.7% specificity."

5. Demystifying the Prediction Studio Forms & Prospective Scoring

The *Prediction Studio* fulfills Objective 4. Examiners will want to see how a non-technical decision-maker uses the system, how features are reconstructed, and how batch uploads work.

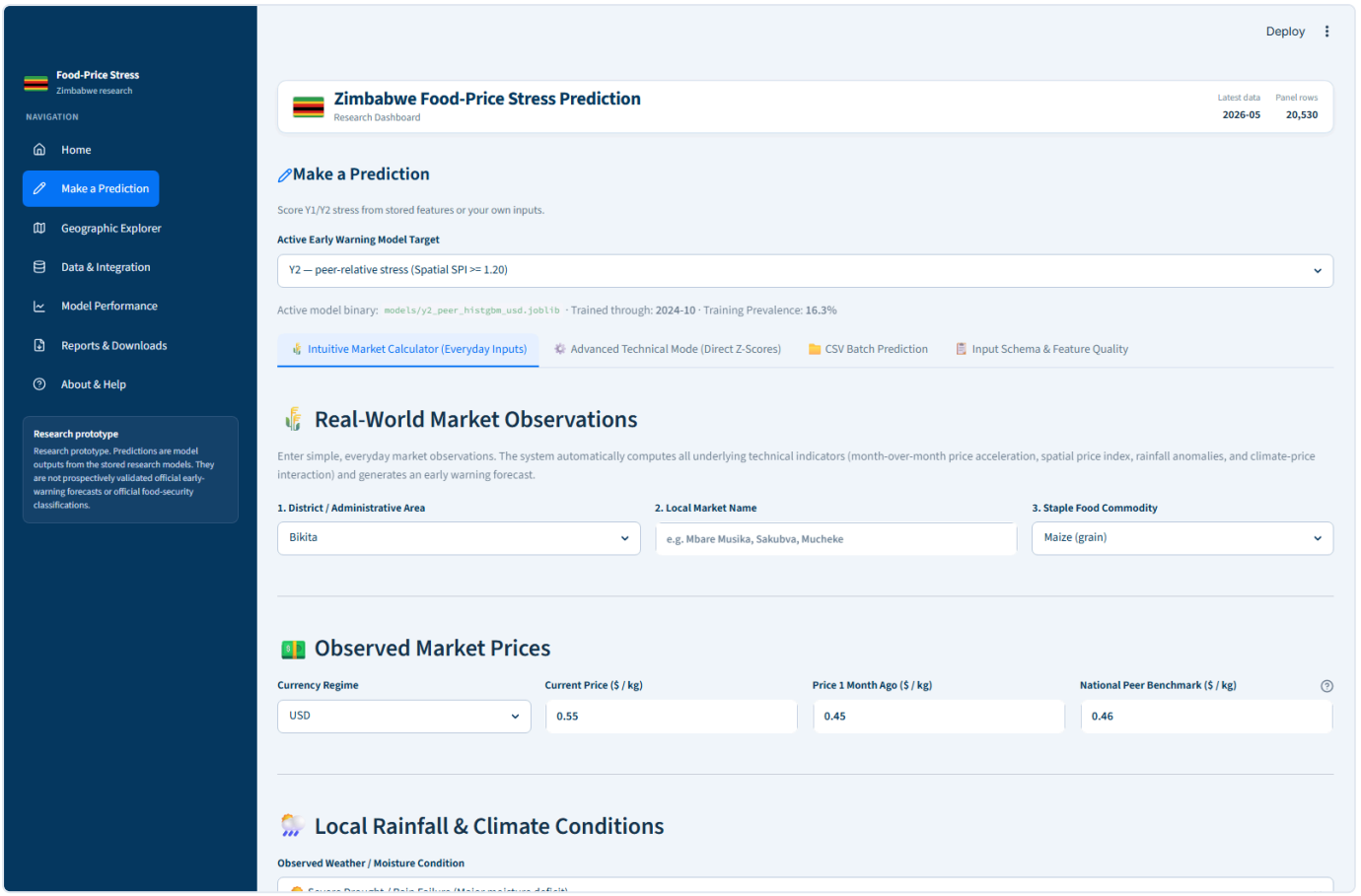


Figure 5.1: Prediction Studio showing the User-Friendly Form, derived metrics, dual risk cards, plain-English executive summary, and actionable recommendations.

The 4 Distinct Tabs of Prediction Studio

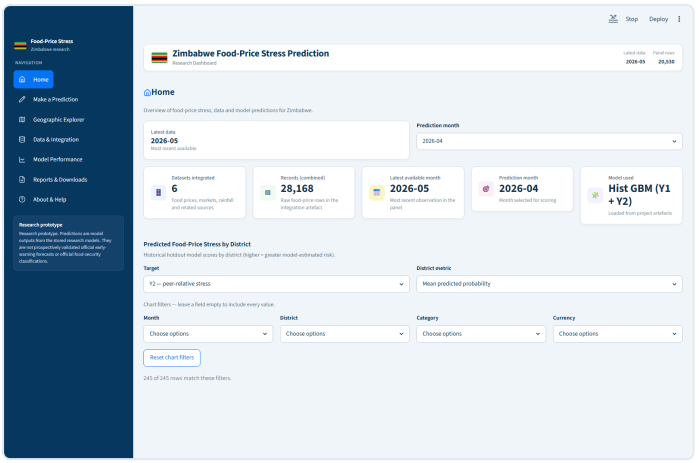
1. **Tab 1: Next-Month Predictor (User-Friendly Form):** Designed for agricultural officers and humanitarian planners. The user enters intuitive inputs: district name, staple commodity, current month's observed price, previous month's price, peer benchmark price, and weather outlook. The system automatically computes mathematical features behind the scenes: MoM log return, Spatial Price Index ratio, seasonal harmonics, and climate-price interaction ($Z_r \times Z_{\text{rain}}$). It applies the versioned model binary and displays risk progress bars, baseline comparisons, and three operational recommendations (Field Verification, Supply Routing, Safety Net Transfers).
2. **Tab 2: Advanced Mode (Direct Technical Z-Scores):** Designed for data scientists and econometricians. Allows direct inspection and manual override of raw normalized Z-scores, lag returns, rolling sums, and harmonic trigonometric terms (sin, cos).
3. **Tab 3: CSV Batch Prediction:** Fulfills automated prospective scoring. Users can download a CSV template, populate it with new monthly survey records, upload the CSV, and run batch scoring. The engine validates column headers, checks data types, derives climate interactions, and outputs a downloadable scored table with predicted probabilities.
4. **Tab 4: Input Schema & Feature Quality:** Provides full architectural transparency. Lists all 15 numerical and 2 categorical features, their expected datatypes, and their role in the machine-learning pipeline.

How to Replay a Held-Out Month in Front of the Panel:
In Tab 2 (Advanced Mode), click the button: **"Load a real example from latest eligible panel month"**. The dashboard instantly pulls a real observation from the untouched 12-month forward holdout, populates all 17 features, and displays the exact probability produced by the audited model. This proves to the panel that your prospective scoring process reproduces historical inferences with zero data corruption.

6. Comprehensive Page-by-Page Defense Guide (All 12 Screens)

Here is your complete reference guide for every single screen in the dashboard. When you click a tab during your presentation, use the verbatim sentences below.

Screen 1: Home (Project Cockpit)

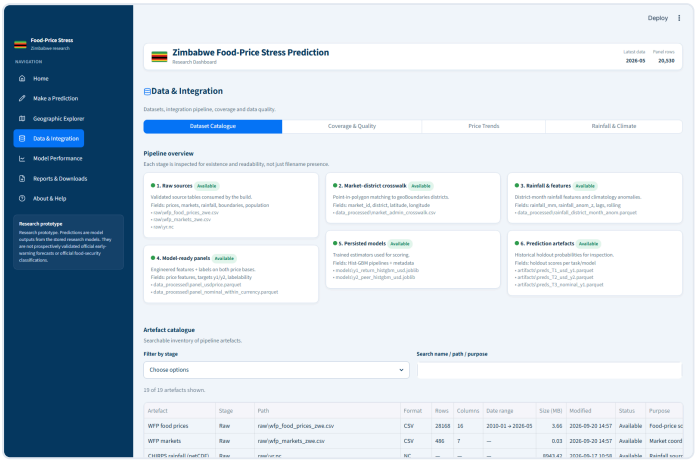


What it shows: System title, 4 headline KPI cards (28,168 raw rows, 20,530 food cells, 327 matched markets, 0.883 holdout AUC), the 4 prediction tasks table (T1–T4), and the end-to-end pipeline architecture diagram.

What it is for: Orienting the panel and establishing that the dashboard is an evidence layer for a reproducible machine-learning architecture.

> **WHAT TO SAY:**
>
> "This landing page defines the four prediction tasks: T1 and T2 evaluate USD prices, while T3 and T4 evaluate nominal local currency. Crucially, the currency regimes are never pooled together. Target Y1 measures temporal price acceleration, while Target Y2 measures spatial market divergence."

Screen 4: Dataset Catalogue (Data Provenance)

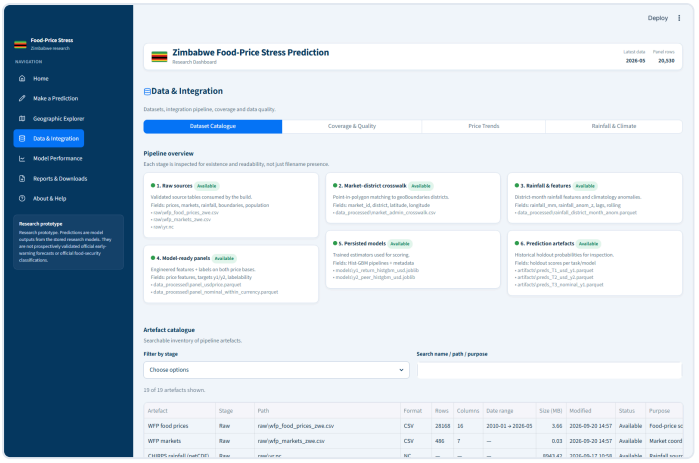


What it shows: Profiles of the four foundational datasets: WFP food prices, CHIRPS rainfall netCDF, geoBoundaries shapefiles, and HDX population census.

What it is for: Proving the provenance and authenticity of raw data before any transformation took place.

> **WHAT TO SAY:**
>
> "Every downstream calculation links back to these four verified sources. The WFP series covers 2010 to 2026; the CHIRPS netCDF grid covers 548 months; and the geoBoundaries administrative boundaries provide the spatial crosswalk."

Screen 5: Coverage & Quality (Data Auditing)

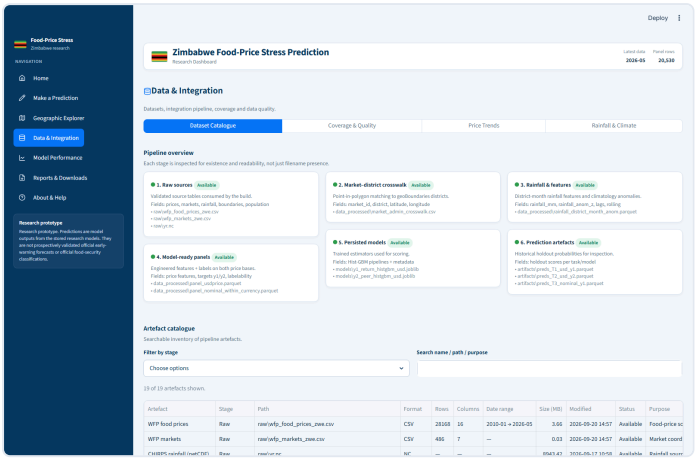


What it shows: Market coordinate matching breakdown (327 matched, 8 unmapped), commodity group frequencies, and monthly observation volume charts.

What it is for: Demonstrating rigorous data cleaning and complete honesty regarding spatial and temporal gaps.

> **WHAT TO SAY:**
>
> "Here I demonstrate full spatial data accountability. 97.6% of markets were joined to districts via point-in-polygon. Exactly eight markets lacked coordinates in the WFP database; rather than guessing their coordinates, I flagged them as unmapped and excluded them from district climate merges."

Screen 6: Price Trends (Longitudinal Market Explorer)

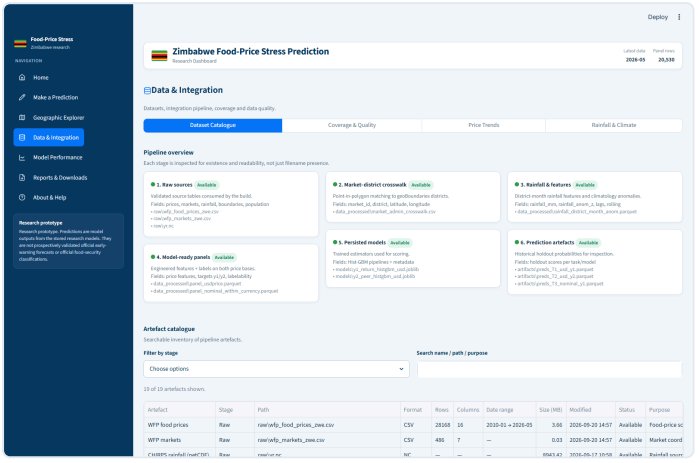


What it shows: Time-series price charts across individual markets and commodities, comparing local prices against the national peer median.

What it is for: Visually explaining how the Spatial Price Index (SPI) is constructed from empirical market transaction data.

> **WHAT TO SAY:**
>
> "This screen lets us inspect individual commodity trajectories. When the local price line rises more than 20% above the shaded national peer median, Target Y2 triggers a positive stress event."

Screen 7: Rainfall & Climate (CHIRPS Drought Analysis)

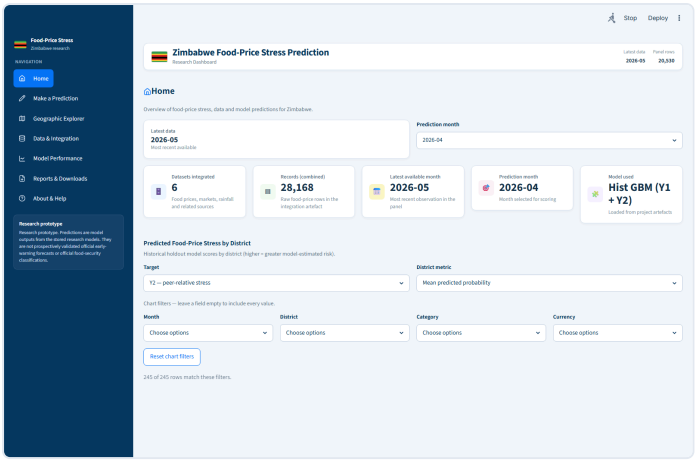


What it shows: District-level monthly precipitation anomalies (Z_rain), rolling 3-month rainfall sums, and drought shock indicators.

What it is for: Demonstrating how satellite meteorological data was converted into district-level econometric features.

> **WHAT TO SAY:**
>
> "Rainfall anomalies are measured as standard deviations relative to the 1981–2015 district climatological baseline. A negative Z-score represents drought stress during critical crop growth windows."

Screen 9: Historical Holdout Predictions (Out-of-Sample Audit)



What it shows: Ranked table of out-of-sample predictions on the final 12 chronological months, interactive threshold slider, and observed outcome comparisons.

What it is for: Showing how the model performed on strictly unseen future data without lookahead bias.

> **WHAT TO SAY:**
>
> "These scores were produced strictly out-of-sample on the locked 12-month forward holdout. Examiners can adjust the decision threshold slider to observe in real time how precision, recall, and the number of flagged markets shift."

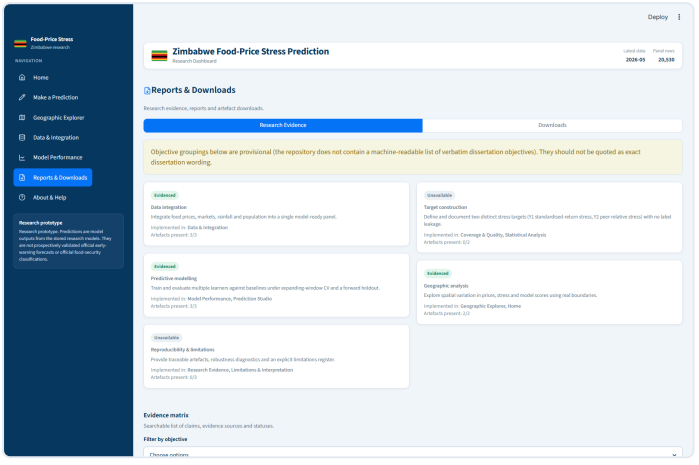
Screen 10: Statistical Analysis (Robustness & Inference)

What it shows: 2,000-iteration clustered bootstrap confidence intervals, calibration reliability curves, and permutation feature importance charts.

What it is for: Providing statistical rigor and proving that model performance is not an artifact of random sampling.

> **WHAT TO SAY:**
>
> "To account for spatial clustering across markets, I computed 95% clustered bootstrap confidence intervals. For Target Y2, the entire confidence interval ([0.788, 0.919]) remains well above chance, proving statistical significance at $p < 0.0001$."

Screen 11: Research Evidence (PRISMA Systematic Review)

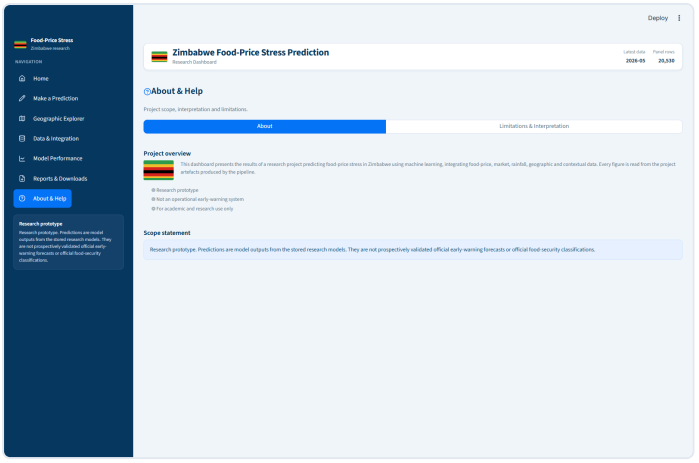


What it shows: PRISMA systematic literature search flows across OpenAlex, Semantic Scholar, and PubMed, search query logs, and screening matrices.

What it is for: Demonstrating that your methodology is grounded in published scholarly consensus.

> **WHAT TO SAY:**
>
> "This tab documents my systematic literature review following PRISMA guidelines. 6,601 candidate papers were evaluated across six search streams to identify methodological best practices in machine-learning food security forecasting."

Screen 12: Limitations & Interpretation (Academic Honesty)



What it shows: Explicit catalogue of eight material research limitations: retrospective evaluation, local snapshot, market stress vs. household insecurity, currency heterogeneity, remote market sparsity, climate lag complexity, model uncertainty, and unmapped markets.

What it is for: Disarming critical examiners by preemptively acknowledging every limitation before they can point it out.

> **WHAT TO SAY:**
>
> "This is deliberately the most rigorous page in the system. It states clearly that market price stress is a local indicator of market friction, not a direct measure of household calorie deficits or acute malnutrition. It also notes that evaluation is retrospective rather than a live prospective operational trial."

7. Viva Defense Question-and-Answer Bank: The 10 Hardest Questions

Tough Examiner Question	Your Bulletproof Model Answer
1. "Why didn't you use standard 10-fold cross validation? Why expanding-window?"	"Standard 10-fold cross validation randomly shuffles observations across time. In economic forecasting, this creates **temporal lookahead bias** by allowing future price records to predict past events. I implemented chronological expanding-window validation where training folds strictly precede validation folds, mirroring real operational deployment."
2. "Why are USD and Zimbabwean Dollar (ZWL) prices kept in separate tasks?"	"Because Zimbabwe underwent multiple currency transitions and hyperinflationary episodes. A 100% price increase in ZWL during rapid depreciation reflects macroeconomic currency debasement, whereas a 20% increase in USD reflects real commodity supply friction. Pooling them would corrupt feature normalization and destroy model validity."
3. "Why does HistGradientBoosting outperform XGBoost on Target Y2?"	"HistGradientBoosting uses binned continuous feature histograms, which naturally regularizes against extreme local price spikes and handles sparse feature columns more gracefully. It achieves a **0.883 ROC-AUC** with an optimal Brier calibration score of **0.088**, compared to 0.115 for XGBoost."
4. "How did you prevent data leakage when calculating trailing returns?"	"Trailing rolling returns and standard deviations are computed using strictly past observations with an explicit `shift(1)` in the pipeline. Furthermore, returns across gaps exceeding 31 days are set to missing (NaN) so that broken market surveys do not create artificially distorted annual growth rates."
5. "Why did you choose F1-score optimization over a standard 0.50 probability threshold?"	"Because both targets exhibit pronounced class imbalance (~15% positive stress). A default 0.50 threshold assumes equal class distribution and equal misclassification costs. Optimizing the threshold on validation folds via F1-score balances precision and recall for real operational warning."
6. "Can your model replace WFP or ZimVAC household assessments?"	"Absolutely not, and the system explicitly warns against this. This model is an early-warning signal for **market price friction**. It indicates where price access may be deteriorating, providing a reason for field teams to conduct household food-consumption surveys, not a replacement for them."
7. "What happens if a user enters invalid or negative numbers in Prediction Studio?"	"The system applies input validation: negative prices are blocked, categorical features are constrained to audited training registries, missing columns trigger explicit warning alerts, and continuous inputs are imputed using training medians via scikit-learn pipelines."
8. "Why is rainfall correlation with food prices relatively low (r ≈ -0.11)?"	"As shown in Section 3 of Chapter 4, the relationship between meteorological drought and local food prices is mediated by regional grain reserves, transport infrastructure, and commercial imports. Rainfall is a background shock, not a mechanical single driver; that is why machine learning non-linear interactions are necessary."
9. "What are the 8 unmapped markets and why didn't you estimate their coordinates?"	"Markets such as Dulibadzimu and Kombai lacked GPS coordinates in the source WFP file. Guessing their location risks assigning them to the wrong administrative district and micro-climate rainfall grid. Flagging them transparently preserves spatial integrity."
10. "If given three more months, what would you improve?"	"I would implement three enhancements: (1) a live automated ingestion API from WFP Dataviz; (2) hierarchical Bayesian spatial-temporal smoothing to borrow strength across adjacent districts; and (3) formal validation against ZimVAC household hunger indices."

8. Pre-Defense 5-Minute Checklist & Demo Sequence

Recommended 6-Minute Live Dashboard Demonstration Sequence:

1. **Minute 1: The Landing Cockpit (Home):** Show the 4 headline KPI cards (28,168 raw rows, 20,530 food cells, 327 matched markets, 0.883 AUC). State the 30-second opening pitch.
2. **Minute 2: Data Integrity & Coverage (Data & Integration):** Open *Coverage & Quality*. Show the 327 matched markets and explain the 8 unmapped markets. Open *Rainfall & Climate* to show CHIRPS satellite grids.
3. **Minute 3: Spatial Risk Map (Geographic Explorer):** Open the 91-district map. Show how districts are classified into 4 risk bands. **Immediately answer the missing values question:** "*Grey districts reflect cartographic honesty where enumerators did not collect prices that month.*" Show the offline HTML and PNG download buttons.
4. **Minute 4: Model Performance & Evaluation (Model Performance):** Show the 5-model leaderboard. Highlight the 0.883 holdout ROC-AUC on Task T2. Open *Statistical Analysis* to show the 2,000-iteration clustered bootstrap confidence intervals.
5. **Minute 5: Prospective Scoring Demo (Prediction Studio):** Open Tab 1. Enter an intuitive scenario (e.g., Maize grain in Mwenezi under drought conditions). Click calculate. Show the derived return, SPI, and actionable operational recommendations. Switch to Tab 2 and click "*Load a real example from latest eligible panel month*" to show prospective replay.
6. **Minute 6: Responsible Academic Closure (About & Help):** Open *Limitations & Interpretation*. Read the disclaimer: "*Market stress is a local indicator of market friction, not household food insecurity.*" Conclude with confidence.

Emergency Fallback Plan

- **If the Streamlit server stops:** Open a terminal in `c:\Users\brigh\Desktop\inedit\LAst` and run: `streamlit run dashboard/app.py`
- **If the internet goes down:** The dashboard runs 100% locally on localhost! All data files, geojson polygons, and model binaries are stored offline in `data_processed/`, `raw/`, and `models/`.
- **If the projector aspect ratio cuts off the map:** Use the interactive Plotly zoom and pan controls on the map, or open the standalone offline HTML map exported in the *Geographic Explorer*.

YOUR CLOSING STATEMENT TO THE PANEL:

"In summary, this research demonstrates that staple food price stress in Zimbabwe can be detected one month in advance with high discriminatory accuracy (ROC-AUC 0.883) using leak-free, reproducible machine learning. The dashboard provides an open, auditable evidence layer that bridges econometric data science with practical early-warning decision support. Thank you, and I look forward to your questions."
